



**SCHOOL OF ELECTRONICS ENGINEERING**  
**CONTINUOUS ASSESSMENT TEST - I**  
**WINTER SEMESTER 2024-2025**

**SLOT: C1 +TC1+TCC1**

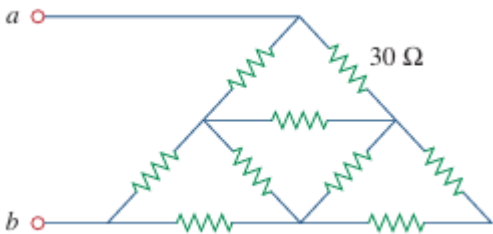
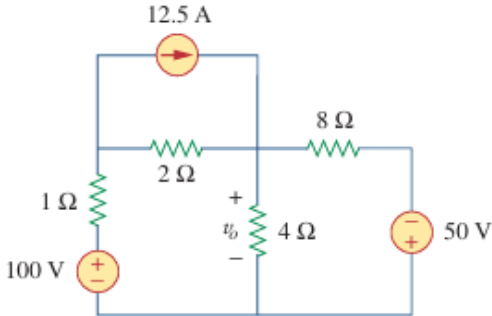
Programme Name & Branch : B. Tech BVD  
Course Code and Course Name : BAECE103 – Network Theory  
Faculty Name(s) : Dr.Elizabeth Rufus, Dr. Sathya P  
Class Number(s) : VL2025260105991, VL2025260105992  
Date of Examination : 19.08.2025  
Exam Duration : 90 minutes Maximum Marks: 50

**General instruction(s):**

- Answer All Questions.
- M - Max mark; CO – Course Outcome; BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)
- Course Outcomes:

CO1: Compute DC circuit parameters using circuit laws and theorems.

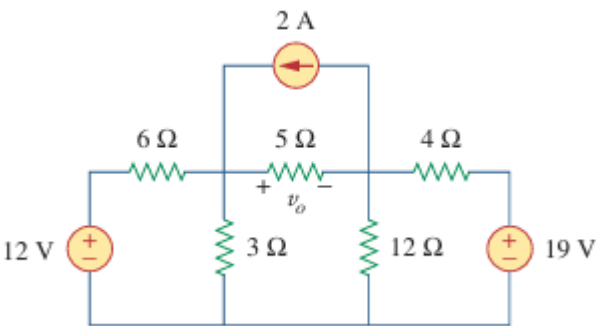
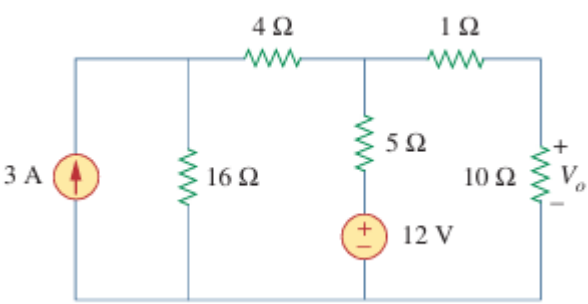
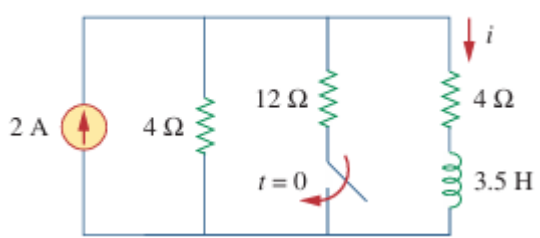
CO2. Determine the natural and step responses of RC, RL, and RLC circuits.

| Q. No | Question                                                                                                                                                                                                | M  | CO | BL |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|
| 1.    | Obtain the equivalent resistance in the circuit of Fig. 1. All resistors have a value of $30\ \Omega$ .<br><br>Fig.1 | 10 | 1  | 2  |
| 2.    | Using nodal analysis, find in the circuit of Fig.2<br><br>Fig.2                                                      | 10 | 1  | 3  |



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|    |                                                                                                                                                                                                                           |    |   |   |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---|---|
| 3. | <p>Determine <math>v_o</math> in the circuit of Fig.3 using the superposition principle.</p>  <p>Fig.3</p>                               | 10 | 1 | 3 |
| 4. | <p>Apply Thevenin's theorem to find <math>V_o</math> in the circuit of Fig.4.</p>  <p>Fig.4</p>                                         | 10 | 1 | 3 |
| 5. | <p>Determine the inductor current for both <math>t &lt; 0</math> and <math>t &gt; 0</math> in the circuits of Fig.5.</p>  <p>Fig.5</p> | 10 | 2 | 3 |